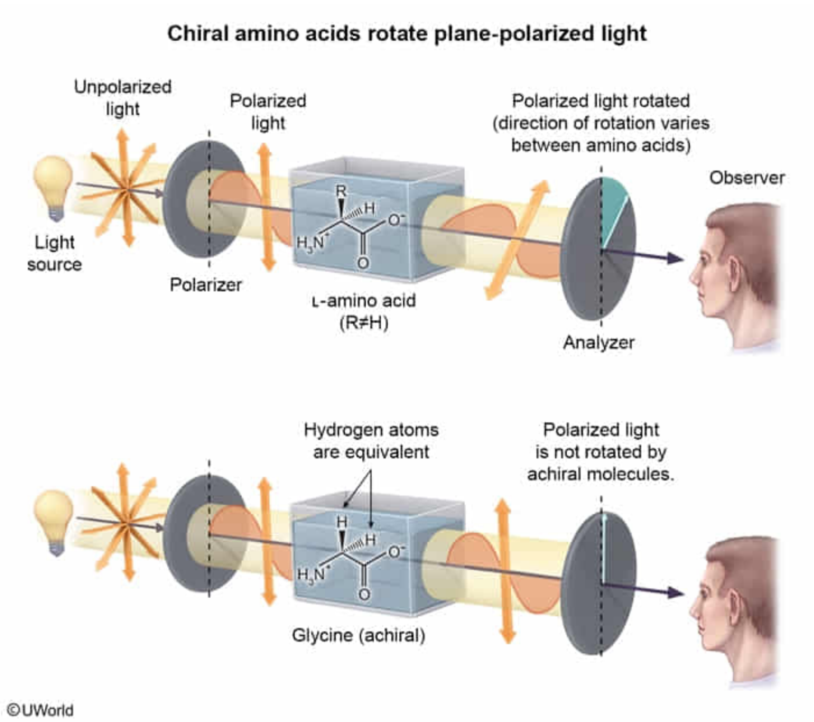
A solution of which amino acid will NOT rotate plane polarized light?

- A. Lysine [1%]
- ✔ B. Glycine [93%]
- C. Tyrosine [3%]
- D. Serine [1%]

Correct
93% answered correctly

Explanation:



Molecules with at least one atom bonded to **four different chemical groups** (a stereocenter) are said to be **chiral**. The bonds to any stereocenter can be arranged in two distinct ways, yielding two possible versions of each stereocenter in a chiral molecule. Solutions of chiral molecules **rotate plane polarized light**, and chiral molecules with opposite arrangements at every stereocenter (enantiomers) rotate light by the same amount but in opposite directions.

Amino acids contain an amino group and a carboxylate group linked together through an alkyl carbon, known as the alpha carbon. The alpha carbon is also bonded to a hydrogen atom and to one other chemical group, called the R-group or the side chain. If the **side chain is different** than the other three substituents, the **amino acid is chiral** and is given an L- or D-designation. L-amino acids predominate in nature.

In biological systems, the side chain can be one of 20 chemical groups associated with the standard amino acids. The side chain of glycine is a hydrogen atom, so the alpha carbon of glycine is bonded to two hydrogen atoms. Because the two hydrogen atoms are identical, the glycine alpha carbon is not bonded to four distinct chemical groups. Therefore, glycine is achiral (neither L nor D) and will not rotate plane polarized light.

(Choices A, C, and D) Apart from glycine, all of the standard amino acids have alpha carbons bonded to four unique substituents. Therefore, they are chiral and will rotate plane polarized light.

Educational objective:
Molecules with one or more stereocenters (chiral molecules) rotate plane polarized light. A stereocenter forms when one atom is bonded to four unique groups, and can be arranged in either of two ways. All of the standard amino acids are chiral except glycine.

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